Advanced Algorithms, Homework TODO

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While in this class, you are expected to work on the problems accompanying the textbook (focusing on the odd-numbered exercises), as well as complete any in-class exercises that are not completed during the class period. Below asks you to reflect on your experience working on these problems. The 'hw period' referred to below is from the date the last hw was due until the date this hw is due (tpyically, a two-week span).

1. What problems did you work on during this hw period, and how much time did you spend working on them? If possible, break the reported time down per problem.

   Answer the problems I worked on during this hw period were all of those on the Masters Theorem Handout we worked on in class. Aswell as the towers of hanoi exercises in the text book. Exercise1.1 - 20 mins, Exercise1.3 15 mins, Exercise 1.5 30mins, Exercise 1.7, 25 mins

2. Who did you work with on these problems, and describe the collaboration.

   Answer I worked with Henry, John, and Joey on all of the problems in class this week. Our collaboration went well, and we were able to get pretty far in our exercises in class. We were making sure we were all on the same page the way througout a problem

3. What external resources did you use while working on these problems? (As a reminder, it is highly encouraged to only use external resources vetted by the instructor. You have all of the tools you need to solve these problems without external resources).

   Answer I used the class reading, and my notes from our lectures

4. What went well while working on these problems?

   Answer The aspects of these problems that went well was writing psuedo-code to help better understand runtime analysis and recursion. Getting more comfortable with induction helped with working on these problems

5. What parts of the homework were most challenging?

   Answer The most challenging but also rewarding parts were figuring out how to use induction to prove runtimes of algorithms

6. How does this material relate to 'real-world' applications?

Answer This material relates to real-world applications because induction and recursion are all about breaking down bigger problems into smaller repeatable steps to understand the larger problem at hand. It also relates to finding the most efficient way to get something done when discussing runtime analysis